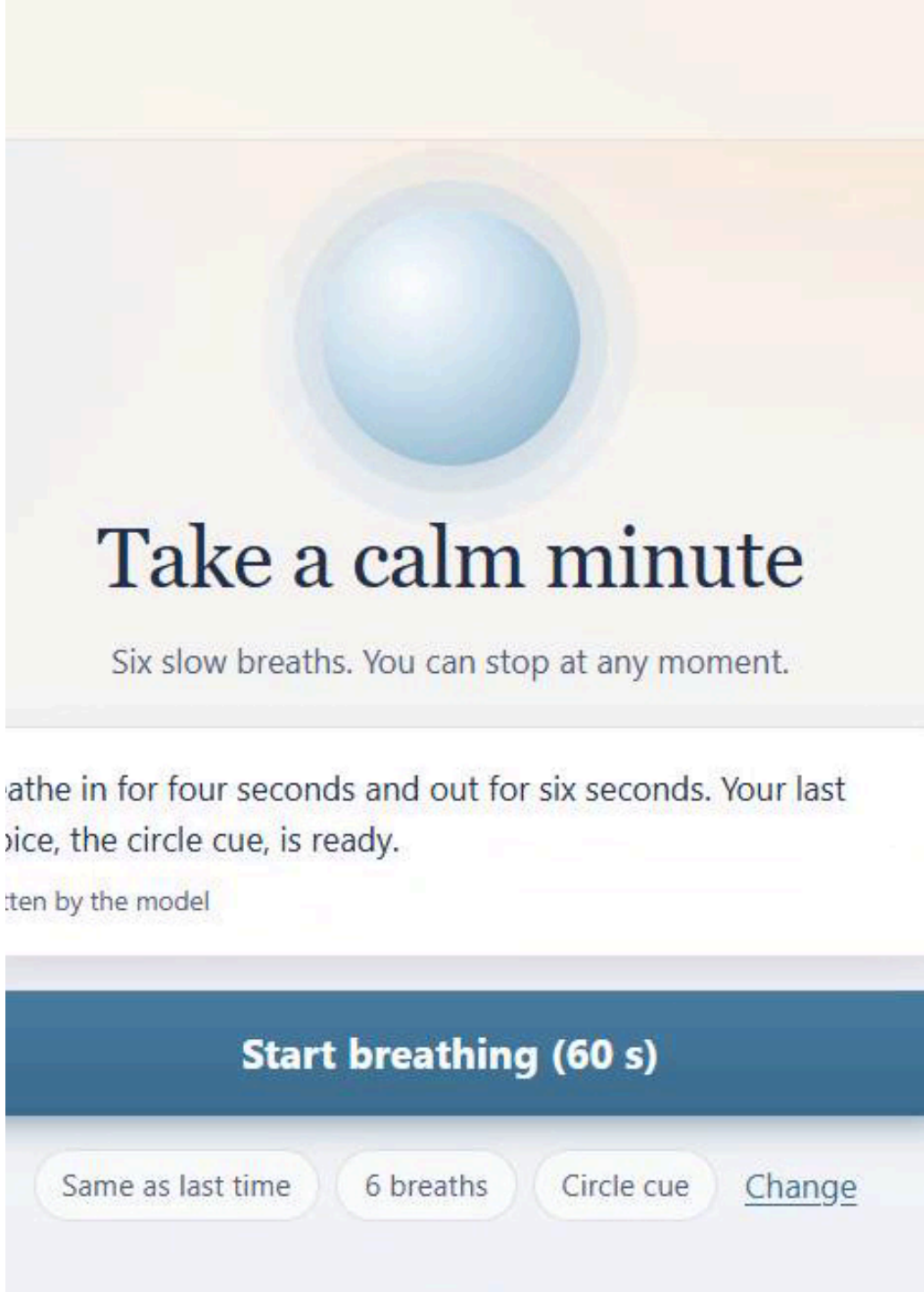
Breathe 60

A calm minute, one skill at a time

A coursework prototype for paced breathing: 4 seconds in, 6 seconds out.

Kelvin Asiedu · IST 402 · Assignment 3



Meet Alex (fictional)

A student with a few minutes between classes who wants to slow down before the next thing, without reading, typing or setting anything up.

1

skill, one button to start

60 s

six breaths, measured by the app

0

questions before you begin

> A class project, not a support service. Nobody is monitoring this app.

The skill, start to finish

1. **Start**

   One button, or press Enter. Nothing to fill in first.

2. **Breathe**

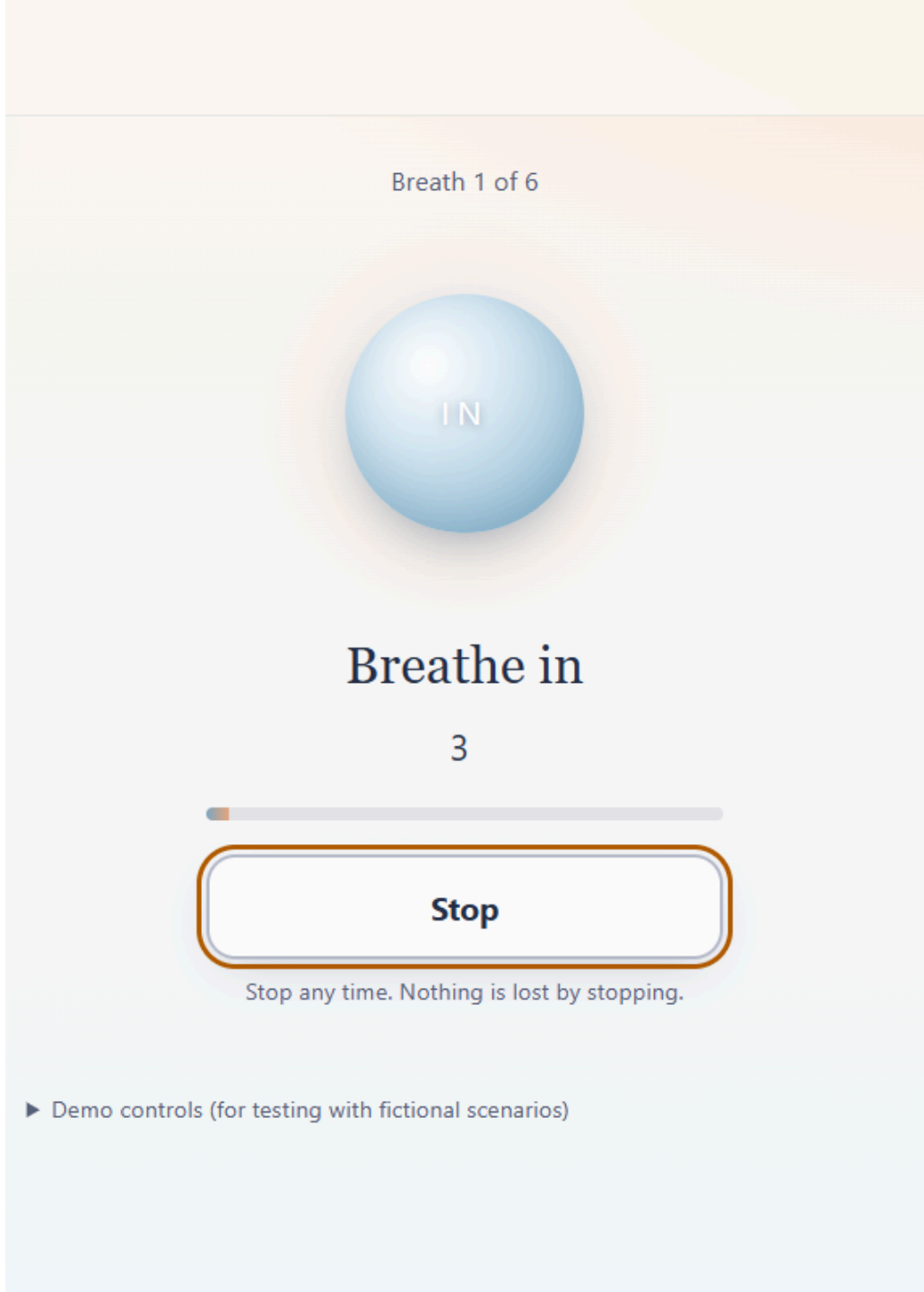
   The orb grows for 4 s and shrinks for 6 s. The pace is carried for you.

3. **Pause for Support**

   Opens from any screen and pauses the exercise.

4. **Stop any time**

   No "are you sure", no streak, no penalty.



Why it looks like this

Each choice comes from the study's findings (15 patients, 13 clinicians).

R1 · Under 90 seconds

People drift after 90 to 100 s. Six breaths take 60 s, measured.

R2 + D4 · Support never moves

In the top bar on every screen, even mid-exercise.

R3 · Permission to stop

Stop is one press and leaving costs nothing.

D2 + D3 · One thing, low effort

One action per screen. No typing to do the skill.

D6 · Calm, not cheerful

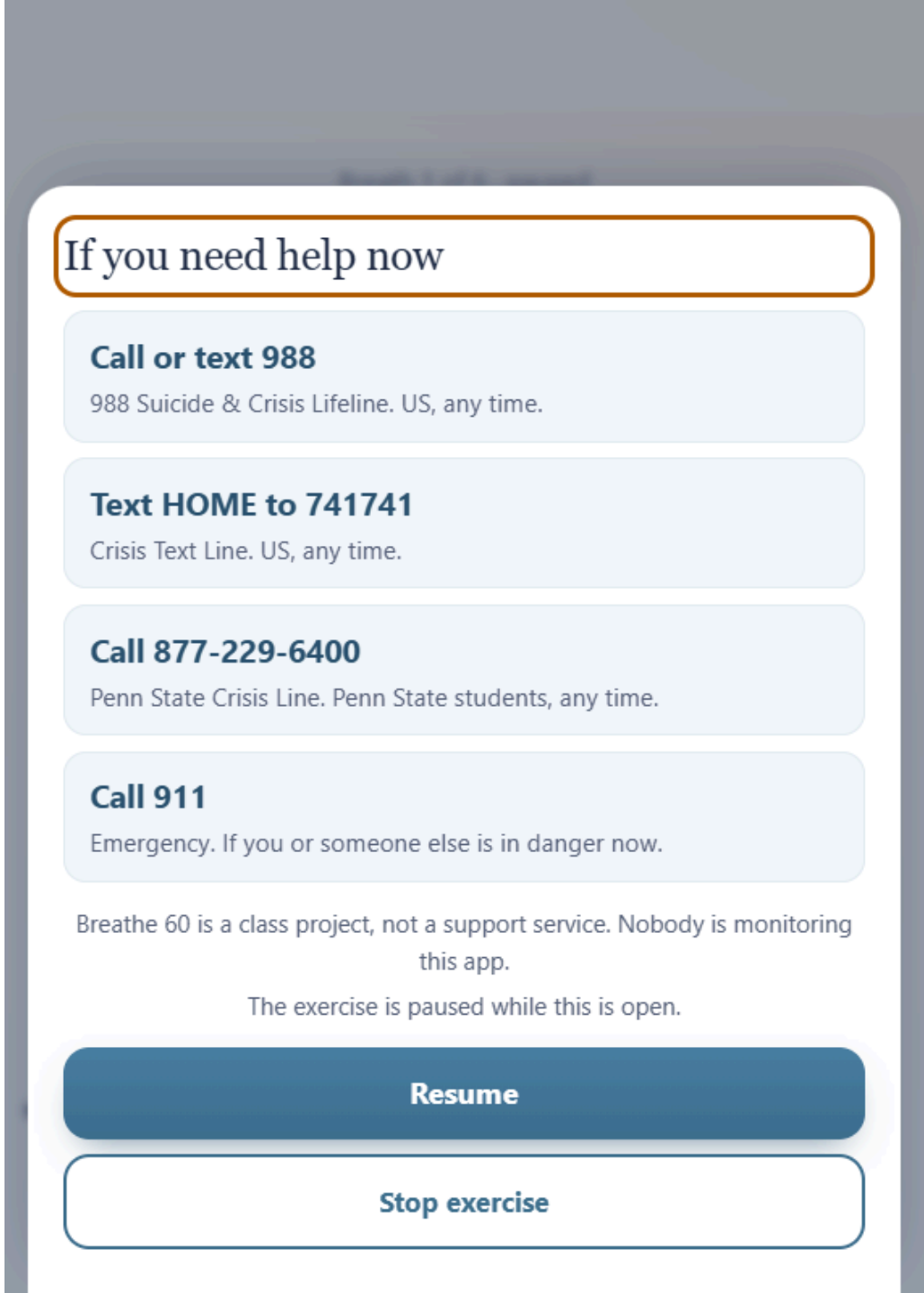
Soft colors, slow motion, plain words, no praise.

R4 · Alex decides

Options are offered, never forced.

Support is always one tap away

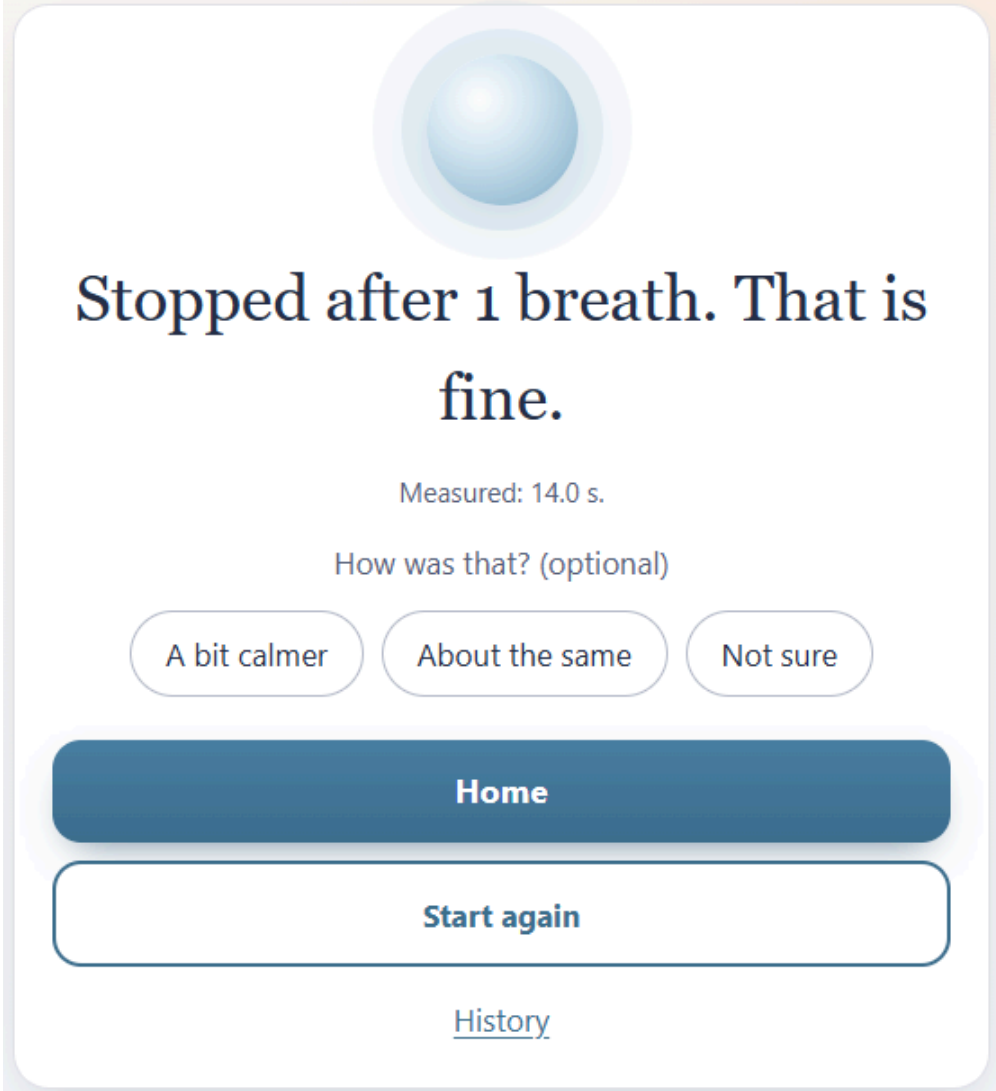
- In the top bar of **every** screen, including inside the exercise
- Static content that ships with the page: works when the AI or the backend is down
- Pauses the exercise until Alex chooses Resume
- Numbers checked against official sources: 988, Crisis Text Line, Penn State Crisis Line, 911
- Opening it is **not** recorded, so nobody is watching



Stopping costs nothing

Stop works at every moment. The app says so plainly and moves on.

- Saved as **stopped**, with the breaths done
- Never counted against Alex, never a streak to break
- Start again is one tap



User state: Alex says, the app adapts

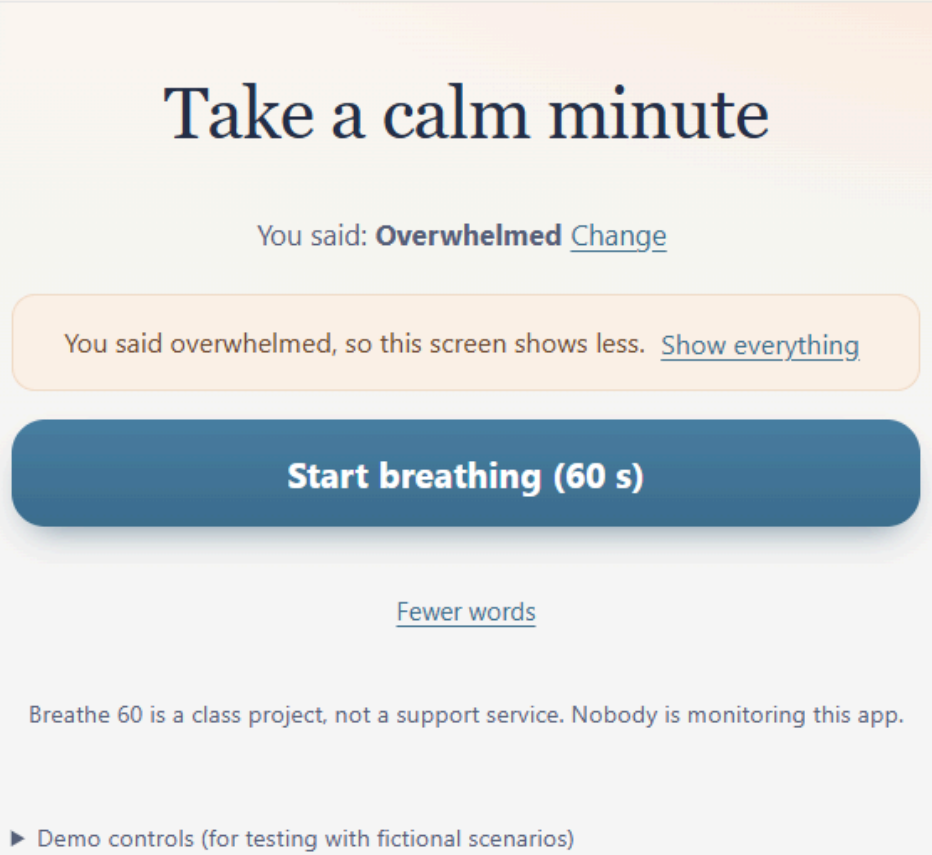
An optional one-tap check-in on Home: **"How are you arriving right now?"** Steady, A bit tense, or Overwhelmed. It is Alex's own word; the app never guesses.

1. **Overwhelmed**
   The screen contracts to the title, Start and Support (D1). **Show everything** undoes it.

2. **A bit tense**
   Length is set to 6 breaths, the shortest (R1). **Use 8 instead** overrides it.

3. **Steady**
   Offers the longer 8 breaths. Ignoring the offer is fine.

> Assumptions: the check-in is optional, never blocks Start or Support, only shapes Home for 6 hours, and "Change" corrects it in one tap. No clinical detector, no sensors, no reading of clicks.

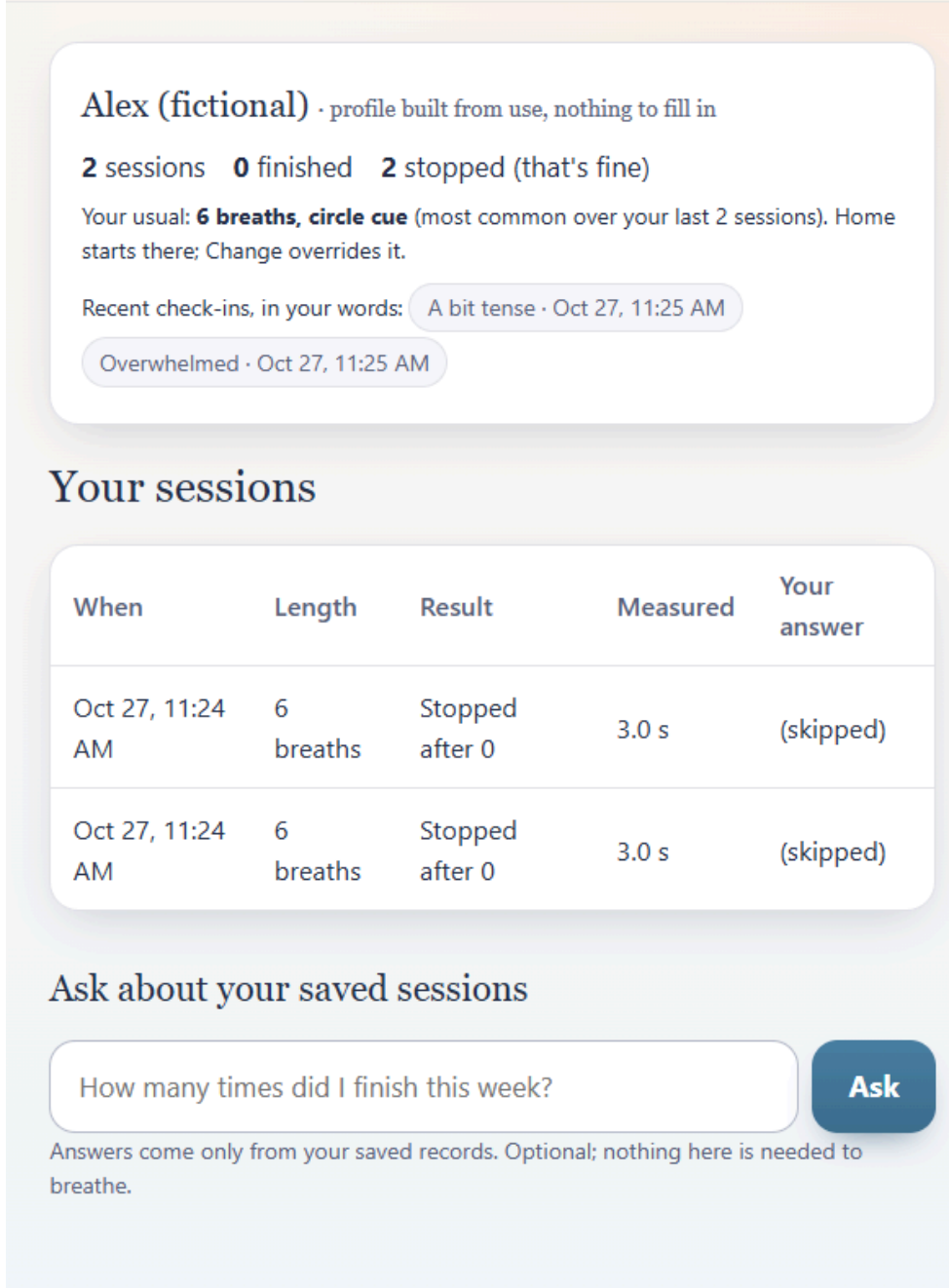


Alex's profile builds up from use

Nothing to fill in. The profile comes from what Alex does:

- **Your usual:** the most common length and cue over the last 10 sessions. Home starts there and shows a "Your usual" pill; **Change** overrides it.
- **Updated every time:** pressing Start saves the chosen length and cue; finishing or stopping saves the result and measured time.
- **Check-ins** and the optional "How was that?" answer are saved as Alex's own words, never as a score.
- **Not saved:** opening Support, any guessed mood.

Accessible for every age: large buttons, high contrast, Enter starts, Escape stops, a motion-free text cue, and it fits a phone screen. No typing is needed to do the skill.



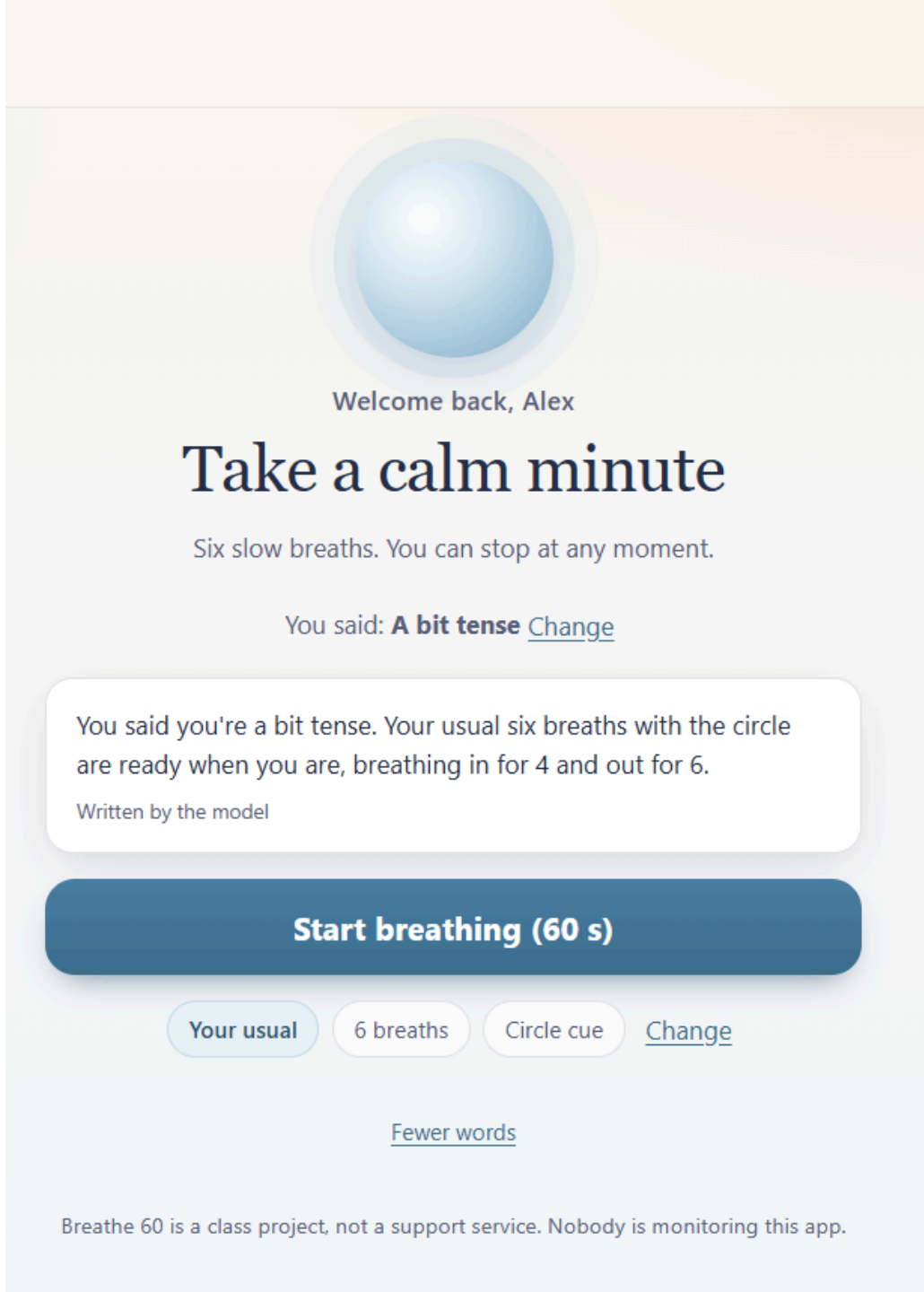
The chatbot: tailored, honest, limited

Tailored by the profile and history. Live, after Alex checked in as "a bit tense":

> "You said you're a bit tense. Your usual six breaths with the circle are ready when you are, breathing in for 4 and out for 6."

Asked about history, it answers only from Alex's own saved rows: *"You've done 2 sessions. Your usual breaths setting is 6."*

Limits: it never runs the exercise, never diagnoses or promises a feeling will change, no praise or clinical words. Asked "Did breathing fix my anxiety?", it said the sessions can't answer that. A rule-breaking reply is replaced with fixed text. The app always says it's a class project.



When something goes wrong

Crisis wording

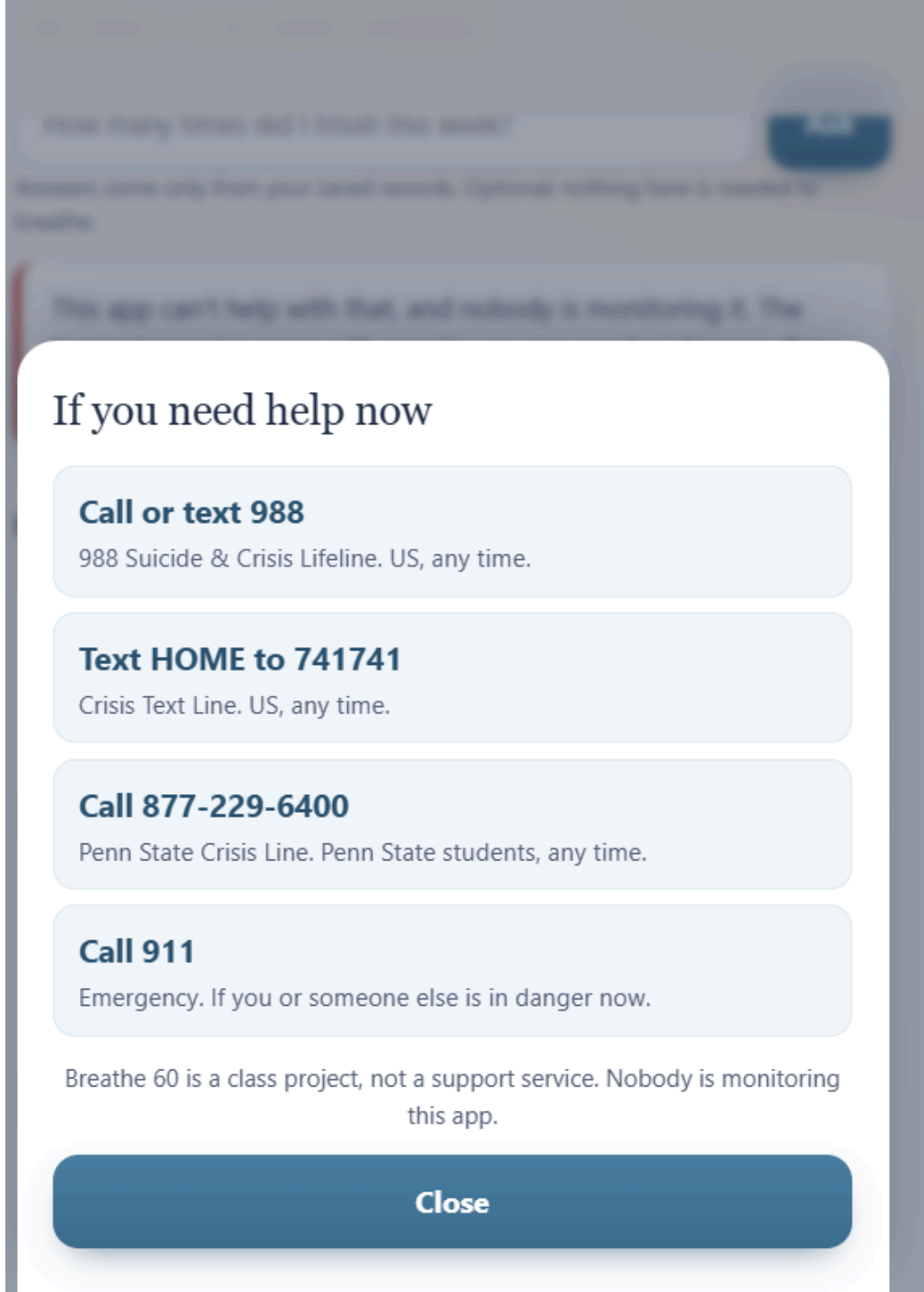
Never reaches the model. The app opens Support with the numbers. A word list can miss things, which is why Support is on every screen anyway.

AI is down

Gemini timed out this morning. The app showed fixed text, said the question could not be answered, and Start and Support kept working. Now it also falls back to another model automatically.

Backend is down

The exercise still runs, and the app says honestly that the session was not saved.



What's real, and what's next

Real and working

- The exercise, Stop and Support
- Saved sessions and history (SQLite)
- AI introduction and questions (Gemini)
- Check-in, usual and profile from history
- 33 backend checks and 13 browser checks pass

Test-only and next

- Mock and Fail modes, for testing failures
- Part 3.3: fifteen fictional scenarios and three behavior checks
- More personalization from history

Thank you. Questions?